An Analytical First-Order Mixed-Zonal Theory for GEqOE

Near-Identity Averaging with Direct Residue Decomposition

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March 11, 2026

Abstract

This note presents a complete analytical first-order theory for the Generalized Equinoctial Orbital Elements (GEqOE) under autonomous Earth zonal harmonics through J_5, comprising closed-form mean-element propagation, closed-form short-period corrections, and closed-form osculating reconstruction. Three main results are established. First, the GEqOE state vector admits an exact first-order orbit-averaging operator in the generalized eccentric longitude K, yielding a finite Fourier-harmonic mean-element system in the single slow angle $\omega = \Psi - \Omega$. For pure J_2, the secular closure is especially simple: ν, g, and Q are constant, while the equinoctial pairs (p_1, p_2) and (q_1, q_2) undergo uniform rotations with explicit rates. Second, a direct residue decomposition method is developed for constructing the short-period kernels: by exploiting the fixed pole structure of the dM/dF Jacobian (where $F = e^{if}$ is the complex true anomaly exponential), each kernel reduces to a polynomial-plus-simple-poles form that is free of spurious logarithmic terms and scales gracefully to high zonal degree. The resulting expressions are exact closed-form rational functions of (q, Q), precomputed once by symbolic algebra; at runtime the theory requires only the evaluation of these algebraic formulas—no numerical quadrature, no interpolation, no iterative solution (except the standard Kepler equation). Third, the resulting analytical mean-plus-short-period model is validated numerically against full osculating GEqOE Taylor integration, achieving sub-kilometer osculating position reconstruction for low-eccentricity orbits and few-kilometer reconstruction for high-eccentricity ($e = 0.65$) orbits over 10+ orbital periods, with reconstruction errors scaling linearly with the zonal perturbation parameter as expected for a first-order theory.

1 Introduction

This note develops the averaged formulation for the Generalized Equinoctial Orbital Elements (GEqOE) of Baù et al. [1] under the autonomous conservative case generated by the Earth zonal harmonics. The focus is on the GEqOE state

$$\boldsymbol{y} = (\nu, p_1, p_2, K, q_1, q_2), \tag{1}$$

under the following assumptions:

1. the conservative perturbation is autonomous and axisymmetric about the inertial z axis;
2. the disturbing potential is a static zonal expansion truncated at degree N (in practice $N = 5$);
3. averaging is performed with respect to the propagated fast angle K;
4. only first-order dynamics are derived: all quantities of order $O(J_n^2)$ and all cross-zonal products $O(J_n J_m)$ are neglected.

The first-order near-identity averaging approach follows the Lie–Deprit–Hori perturbation framework [4, 5], adapted to the non-canonical GEqOE variables. The restriction to autonomous axisymmetric perturbations is deliberate. It keeps the averaged problem autonomous and avoids the additional complications introduced by explicit time dependence (third-body ephemerides, rotating-frame geopotential tides, etc.).

The classical analytical treatment of the zonal satellite problem originates with Brouwer [2] and Kozai [3], who derived first-order secular and short-period corrections in Delaunay variables. Subsequent developments by Coffey and Deprit [9] extended the theory to third order, while de Saedeleer [10] provided a generic compact first-order closed-form solution for any individual zonal harmonic. Lara [11] recently revisited the complete Brouwer solution using a single canonical transformation in Poincaré style. In parallel, the Draper Semi-analytical Satellite Theory (DSST) [12, 13] developed the semi-analytical approach in non-singular equinoctial elements [6]. The present work differs from these in that it operates directly in the GEqOE variables of [1]—which are non-canonical but singularity-free—and replaces the Hamiltonian Lie series machinery with a direct residue decomposition in the complex true anomaly variable F.

The note is organized as follows. Sections 2–5 establish the GEqOE dynamics, the averaging operator, and the especially simple J_2 secular closure. Section 6 develops the near-identity transformation theory and the constructive first-order short-period map. Sections 7–8 introduce the complex true anomaly variables and the direct residue decomposition algorithm that makes the short-period kernels tractable through high zonal degree. Section 9 summarizes the operational structure of the resulting analytical theory and clarifies the role of mean-state propagation. Section 10 analyzes the finite-harmonic structure of the general zonal mean drift. Section 11 validates the complete model against osculating GEqOE Taylor integration. Section 12 sketches the path to a Fourier-in-K thrust extension, and Section 13 summarizes the conclusions.

2 GEqOE Preliminaries

Define the auxiliary quantities

$$g^2 = p_1^2 + p_2^2, \qquad \beta = \sqrt{1-g^2}, \qquad \alpha = \frac{1}{1+\beta}, \tag{2}$$

$$a = \left(\frac{\mu}{\nu^2}\right)^{1/3}, \tag{3}$$

$$Q^2 = q_1^2 + q_2^2, \qquad \gamma = 1 + Q^2, \qquad \delta = 1 - Q^2. \tag{4}$$

From the equinoctial orientation variables one has

$$\cos i = \frac{\delta}{\gamma}, \qquad \tan\frac{i}{2} = Q. \tag{5}$$

The propagated generalized eccentric longitude K gives

$$r = a\left(1 - p_1 \sin K - p_2 \cos K\right), \tag{6}$$

$$X = a\left(\alpha p_1 p_2 \sin K + (1 - \alpha p_1^2)\cos K - p_2\right), \tag{7}$$

$$Y = a\left(\alpha p_1 p_2 \cos K + (1 - \alpha p_2^2)\sin K - p_1\right), \tag{8}$$

and the normalized inertial z coordinate is

$$\hat{z} = \frac{2(Y q_2 - X q_1)}{\gamma r}. \tag{9}$$

The generalized angular momentum proxy used throughout [1] and the current implementation is

$$c = \left(\frac{\mu^2}{\nu}\right)^{1/3} \beta. \tag{10}$$

Under purely conservative autonomous dynamics, the generalized energy is a first integral and therefore

$$\dot{\nu} = 0. \tag{11}$$

3 Autonomous Zonal GEqOE Dynamics

For a zonal potential of the form

$$U(r, \hat{z}) = \sum_{n \geq 2} U_n(r, \hat{z}) = \sum_{n \geq 2} \frac{C_n}{r^{n+1}} P_n(\hat{z}), \qquad C_n = \mu J_n R_e^n, \tag{12}$$

the current branch uses the exact conservative GEqOE system

$$\dot{\nu} = 0, \tag{13}$$

$$\dot{p}_1 = p_2(d - w_h) + \frac{1}{c}\left(\frac{X}{a} + 2p_2\right)\mathcal{E}_Z, \tag{14}$$

$$\dot{p}_2 = p_1(w_h - d) - \frac{1}{c}\left(\frac{Y}{a} + 2p_1\right)\mathcal{E}_Z, \tag{15}$$

$$\dot{K} = \frac{w}{r} + d - w_h + \frac{1}{c}\left(1 + \alpha\left(1 - \frac{r}{a}\right)\right)\mathcal{E}_Z, \tag{16}$$

$$\dot{q}_1 = \frac{\gamma}{2} w_Y, \tag{17}$$

$$\dot{q}_2 = \frac{\gamma}{2} w_X, \tag{18}$$

with

$$w = \sqrt{\frac{\mu}{a}}, \tag{19}$$

$$h = \sqrt{c^2 - 2r^2 U}, \tag{20}$$

$$d = \frac{h - c}{r^2}, \tag{21}$$

$$\mathcal{E}_Z = \sum_{n \geq 2} (1 - n) U_n, \tag{22}$$

$$F_h = -\frac{\delta}{\gamma r} \frac{\partial U}{\partial \hat{z}}, \tag{23}$$

$$w_X = \frac{X}{h} F_h, \qquad w_Y = \frac{Y}{h} F_h, \qquad w_h = w_X q_1 - w_Y q_2. \tag{24}$$

This formulation is already structurally suitable for averaging because:

1. the system is autonomous;
2. the fast angle is explicit;
3. $\dot{\nu} = 0$ exactly;
4. all forcing enters through $r(K)$, $X(K)$, $Y(K)$, and $\hat{z}(K)$.

4 First-Order Orbit Averaging in K

4.1 Averaging operator

To first order in the zonal perturbation, the unperturbed fast flow is

$$\dot{K}_0 = \frac{w}{r}, \qquad \frac{dt}{dK} = \frac{r}{w}. \tag{25}$$

Hence the orbital average of any 2π-periodic quantity $f(K)$ is

$$\langle f \rangle_K = \frac{1}{T}\int_0^T f\,dt = \frac{1}{2\pi a}\int_0^{2\pi} r(K)\,f(K)\,dK. \tag{26}$$

Let the slow state be

$$\boldsymbol{z} = (\nu, p_1, p_2, q_1, q_2). \tag{27}$$

Then the exact first-order averaged slow flow of the autonomous zonal problem is

$$\dot{\bar{\nu}} = 0, \tag{28}$$

$$\dot{\bar{p}}_1 = \frac{1}{2\pi\bar{a}}\int_0^{2\pi} \bar{r}\left[\bar{p}_2(\bar{d} - \bar{w}_h) + \frac{1}{\bar{c}}\left(\frac{\bar{X}}{\bar{a}} + 2\bar{p}_2\right)\bar{\mathcal{E}}_Z\right] dK, \tag{29}$$

$$\dot{\bar{p}}_2 = \frac{1}{2\pi\bar{a}}\int_0^{2\pi} \bar{r}\left[\bar{p}_1(\bar{w}_h - \bar{d}) - \frac{1}{\bar{c}}\left(\frac{\bar{Y}}{\bar{a}} + 2\bar{p}_1\right)\bar{\mathcal{E}}_Z\right] dK, \tag{30}$$

$$\dot{\bar{q}}_1 = \frac{1}{2\pi\bar{a}}\int_0^{2\pi} \bar{r}\,\frac{\bar{\gamma}}{2}\,\bar{w}_Y\,dK, \tag{31}$$

$$\dot{\bar{q}}_2 = \frac{1}{2\pi\bar{a}}\int_0^{2\pi} \bar{r}\,\frac{\bar{\gamma}}{2}\,\bar{w}_X\,dK. \tag{32}$$

All barred intermediate quantities are evaluated from the frozen slow state $\bar{\boldsymbol{z}}$ and the fast phase K via the unperturbed relations (6)-(9) and (10).

4.2 Interpretation

Equations (29)-(32) are already a valid first-order mean-element model for any autonomous zonal expansion. They are not yet a closed analytical solution, but they are an exact one-revolution quadrature formula in the variables used by the present branch.

This is the right point of departure because it separates the problem into two layers:

1. a *formal averaging layer*, which is already solved by (26)-(32);
2. a *closure layer*, where one asks whether the integrals can be reduced to explicit analytical expressions.

5 Closed First-Order J_2 Secular Model

5.1 Why J_2 is special

For the pure J_2 potential,

$$U_{J_2} = -\frac{A}{r^3}\left(1 - 3\hat{z}^2\right), \qquad A = \frac{\mu J_2 R_e^2}{2}, \tag{33}$$

the averaged disturbing function is exceptionally simple. Using the standard Keplerian averages

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{1}{a^3\beta^3}, \qquad \left\langle \frac{\sin^2 u}{r^3} \right\rangle = \frac{1}{2a^3\beta^3}, \tag{34}$$

with $u = \omega + f$, one obtains

$$\bar{U}_{J_2} = -\frac{A}{2a^3\beta^3}\left(3\cos^2 i - 1\right) = -\frac{A}{2a^3\beta^3}\left(3\frac{\delta^2}{\gamma^2} - 1\right). \tag{35}$$

Equation (35) depends only on a, β, and i. It has no dependence on the node or pericenter angles. Consequently, at first order in J_2,

$$\dot{\bar{\nu}} = 0, \qquad \dot{\bar{g}} = 0, \qquad \dot{\bar{Q}} = 0. \tag{36}$$

Thus the magnitudes of (p_1, p_2) and (q_1, q_2) are constant and the secular dynamics reduce to pure rotations.

5.2 Secular rates

Let

$$c_i = \cos i = \frac{\delta}{\gamma}, \qquad p = a\beta^2. \tag{37}$$

Within first-order singly averaged J_2 theory, the secular drift of the mean GEqOE angular variables coincides with the classical J_2 secular drift [2, 3] written in the GEqOE angles Ψ and Ω. At first order in J_2, the secular nodal rate is

$$\omega_{\Omega,J_2} = -\frac{3}{2}\,\nu J_2 \left(\frac{R_e}{p}\right)^2 c_i = -\frac{3A}{ca^3\beta^3}\,\frac{\delta}{\gamma}, \tag{38}$$

and the secular rate of the generalized longitude of pericenter is

$$\omega_{\Psi,J_2} = \frac{3}{4}\,\nu J_2 \left(\frac{R_e}{p}\right)^2 \left(5c_i^2 - 2c_i - 1\right) = \frac{3A}{2ca^3\beta^3}\left(5\frac{\delta^2}{\gamma^2} - 2\frac{\delta}{\gamma} - 1\right). \tag{39}$$

Because

$$p_1 = g\sin\Psi, \qquad p_2 = g\cos\Psi, \qquad q_1 = Q\sin\Omega, \qquad q_2 = Q\cos\Omega, \tag{40}$$

the first-order mean GEqOE system for J_2 becomes

$$\dot{\bar{\nu}} = 0, \tag{41}$$
$$\dot{\bar{p}}_1 = \omega_{\Psi,J_2}\,\bar{p}_2, \tag{42}$$
$$\dot{\bar{p}}_2 = -\omega_{\Psi,J_2}\,\bar{p}_1, \tag{43}$$
$$\dot{\bar{q}}_1 = \omega_{\Omega,J_2}\,\bar{q}_2, \tag{44}$$
$$\dot{\bar{q}}_2 = -\omega_{\Omega,J_2}\,\bar{q}_1. \tag{45}$$

Equations (41)-(45) are autonomous and closed. They provide exactly the kind of reduced slow-flow model that is suitable for fast metric computation.

5.3 Closed-form integration

Because the coefficients in (42)-(45) are constant along the secular J_2 flow, the system is explicitly integrable. Define the mean angular variables

$$\bar{\Psi} = \operatorname{atan2}(\bar{p}_1, \bar{p}_2), \qquad \bar{\Omega} = \operatorname{atan2}(\bar{q}_1, \bar{q}_2). \tag{46}$$

Then

$$\dot{\bar{\Psi}} = \omega_{\Psi,J_2}, \qquad \dot{\bar{\Omega}} = \omega_{\Omega,J_2}, \tag{47}$$

so that

$$\bar{\Psi}(t) = \bar{\Psi}_0 + \omega_{\Psi,J_2}(t - t_0), \qquad \bar{\Omega}(t) = \bar{\Omega}_0 + \omega_{\Omega,J_2}(t - t_0). \tag{48}$$

Let

$$\bar{g}_0 = \sqrt{\bar{p}_1(t_0)^2 + \bar{p}_2(t_0)^2}, \qquad \bar{Q}_0 = \sqrt{\bar{q}_1(t_0)^2 + \bar{q}_2(t_0)^2}. \tag{49}$$

The closed secular solution is therefore

$$\bar{\nu}(t) = \bar{\nu}_0, \tag{50}$$
$$\bar{p}_1(t) = \bar{g}_0 \sin\left[\bar{\Psi}_0 + \omega_{\Psi,J_2}(t - t_0)\right], \tag{51}$$
$$\bar{p}_2(t) = \bar{g}_0 \cos\left[\bar{\Psi}_0 + \omega_{\Psi,J_2}(t - t_0)\right], \tag{52}$$
$$\bar{q}_1(t) = \bar{Q}_0 \sin\left[\bar{\Omega}_0 + \omega_{\Omega,J_2}(t - t_0)\right], \tag{53}$$
$$\bar{q}_2(t) = \bar{Q}_0 \cos\left[\bar{\Omega}_0 + \omega_{\Omega,J_2}(t - t_0)\right]. \tag{54}$$

Equivalently, the two equinoctial pairs evolve by constant-speed planar rotations,

$$\begin{bmatrix}\bar{p}_1(t)\\ \bar{p}_2(t)\end{bmatrix} = \begin{bmatrix}\cos\theta_\Psi & \sin\theta_\Psi\\ -\sin\theta_\Psi & \cos\theta_\Psi\end{bmatrix}\begin{bmatrix}\bar{p}_1(t_0)\\ \bar{p}_2(t_0)\end{bmatrix}, \qquad \theta_\Psi = \omega_{\Psi,J_2}(t-t_0), \tag{55}$$

$$\begin{bmatrix}\bar{q}_1(t)\\ \bar{q}_2(t)\end{bmatrix} = \begin{bmatrix}\cos\theta_\Omega & \sin\theta_\Omega\\ -\sin\theta_\Omega & \cos\theta_\Omega\end{bmatrix}\begin{bmatrix}\bar{q}_1(t_0)\\ \bar{q}_2(t_0)\end{bmatrix}, \qquad \theta_\Omega = \omega_{\Omega,J_2}(t-t_0). \tag{56}$$

5.4 Mean longitude / phase

If a mean fast phase is also required, the natural choice is the mean generalized longitude

$$\bar{L} = \bar{M} + \bar{\Psi}. \tag{57}$$

Using the standard first-order J_2 secular rate of the mean anomaly,

$$\dot{\bar{M}} = \nu + \frac{3}{4}\nu J_2\left(\frac{R_e}{a\beta^2}\right)^2 \beta\left(3c_i^2 - 1\right), \tag{58}$$

one obtains

$$\dot{\bar{L}} = \nu + \frac{3}{4}\nu J_2\left(\frac{R_e}{a\beta^2}\right)^2 \left[\beta(3c_i^2-1) + 5c_i^2 - 2c_i - 1\right]. \tag{59}$$

6 Near-Identity Transformation and Short-Period Map

6.1 Near-identity transformation

A complete mean-element formulation requires more than the secular equations. It also requires a *near-identity transformation* [4, 5] between:

1. the osculating GEqOE of the full zonal dynamics, and
2. the mean GEqOE propagated by the secular model.

Write the exact first-order zonal problem in the form

$$\dot{\boldsymbol{z}} = \varepsilon \boldsymbol{f}_1(\boldsymbol{z}, K) + O(\varepsilon^2), \tag{60}$$

$$\dot{K} = n_0(\boldsymbol{z}, K) + \varepsilon g_1(\boldsymbol{z}, K) + O(\varepsilon^2), \tag{61}$$

where

$$\boldsymbol{z} = (\nu, p_1, p_2, q_1, q_2), \qquad \varepsilon \sim J_n, \qquad n_0(\boldsymbol{z}, K) = \frac{w}{r}. \tag{62}$$

The complete averaged formulation is then sought in the form

$$\boldsymbol{z}_{\text{osc}} = \bar{\boldsymbol{z}} + \varepsilon \boldsymbol{u}_1(\bar{\boldsymbol{z}}, \bar{K}) + O(\varepsilon^2), \tag{63}$$

$$K_{\text{osc}} = \bar{K} + \varepsilon v_1(\bar{\boldsymbol{z}}, \bar{K}) + O(\varepsilon^2), \tag{64}$$

with periodic short-period corrections $\boldsymbol{u}_1$ and v_1.

The usual normalization is

$$\frac{1}{2\pi}\int_0^{2\pi} \boldsymbol{u}_1(\bar{\boldsymbol{z}}, \bar{K})\, d\bar{K} = \boldsymbol{0}, \qquad \frac{1}{2\pi}\int_0^{2\pi} v_1(\bar{\boldsymbol{z}}, \bar{K})\, d\bar{K} = 0, \tag{65}$$

which fixes the decomposition into mean and periodic parts.

The inverse initialization map, used to convert osculating initial conditions to mean initial conditions, follows directly from (63)-(64):

$$\bar{\boldsymbol{z}}(t_0) = \boldsymbol{z}_{\text{osc}}(t_0) - \varepsilon \boldsymbol{u}_1(\boldsymbol{z}_{\text{osc}}(t_0), K_{\text{osc}}(t_0)) + O(\varepsilon^2), \tag{66}$$

$$\bar{K}(t_0) = K_{\text{osc}}(t_0) - \varepsilon v_1(\boldsymbol{z}_{\text{osc}}(t_0), K_{\text{osc}}(t_0)) + O(\varepsilon^2). \tag{67}$$

The $O(\varepsilon^2)$ initialization error incurred by evaluating $\boldsymbol{u}_1$ at the osculating state instead of the (unknown) mean state is consistent with the first-order theory.

6.2 Homological equations

Substituting (63)-(64) into (61) and matching terms at first order gives the homological equation for the slow variables:

$$n_0(\bar{\boldsymbol{z}}, \bar{K}) \frac{\partial \boldsymbol{u}_1}{\partial \bar{K}} = \boldsymbol{f}_1(\bar{\boldsymbol{z}}, \bar{K}) - \bar{\boldsymbol{f}}_1(\bar{\boldsymbol{z}}), \tag{68}$$

where $\bar{\boldsymbol{f}}_1$ is the averaged slow drift. Therefore

$$\frac{\partial \boldsymbol{u}_1}{\partial \bar{K}} = \frac{\bar{r}}{\bar{w}} \left[\boldsymbol{f}_1(\bar{\boldsymbol{z}}, \bar{K}) - \bar{\boldsymbol{f}}_1(\bar{\boldsymbol{z}}) \right]. \tag{69}$$

Once the secular drift is known, the short-period correction of the slow variables is obtained by a direct quadrature in $\bar{K}$:

$$\boldsymbol{u}_1(\bar{\boldsymbol{z}}, \bar{K}) = \int^{\bar{K}} \frac{\bar{r}(\phi)}{\bar{w}} \left[\boldsymbol{f}_1(\bar{\boldsymbol{z}}, \phi) - \bar{\boldsymbol{f}}_1(\bar{\boldsymbol{z}}) \right] d\phi - (\text{constant chosen to enforce (65)}). \tag{70}$$

The phase correction is slightly more delicate because the unperturbed fast rate $n_0 = w/r$ depends on $\bar{K}$. The first-order phase homological equation is

$$n_0 \frac{\partial v_1}{\partial \bar{K}} - \frac{\partial n_0}{\partial \bar{K}} v_1 = g_1 - \bar{g}_1 + \frac{\partial n_0}{\partial \bar{\boldsymbol{z}}} \cdot \boldsymbol{u}_1, \tag{71}$$

which can be rewritten as

$$\frac{\partial}{\partial \bar{K}} \left(\frac{v_1}{n_0} \right) = \frac{g_1 - \bar{g}_1 + \dfrac{\partial n_0}{\partial \bar{\boldsymbol{z}}} \cdot \boldsymbol{u}_1}{n_0^2}. \tag{72}$$

Hence v_1 is also obtained by quadrature, again with the additive constant fixed by the zero-mean normalization in (65).

6.3 Constructive first-order map in anomaly variables

The previous discussion becomes constructive once the zonal forcing is written in variables adapted to the unperturbed ellipse. Introduce

$$G = K - \Psi, \qquad \omega = \Psi - \Omega, \qquad s_i = \sin i = \frac{2Q}{\gamma}, \qquad c_i = \cos i = \frac{\delta}{\gamma}. \tag{73}$$

Then the unperturbed geometry is

$$r = a(1 - g \cos G), \tag{74}$$

$$\cos f = \frac{\cos G - g}{1 - g \cos G}, \tag{75}$$

$$\sin f = \frac{\beta \sin G}{1 - g \cos G}, \tag{76}$$

$$u = \omega + f, \tag{77}$$

and therefore

$$\hat{z} = s_i \sin u. \tag{78}$$

To first order in J_2 one may set

$$h = c + O(J_2), \qquad d = -\frac{U}{c} + O(J_2^2), \qquad w_h = \frac{3A}{cr^3} c_i \hat{z}^2 + O(J_2^2), \tag{79}$$

with

$$U = -\frac{A}{r^3} \left(1 - 3 s_i^2 \sin^2 u \right). \tag{80}$$

Now parameterize the slow variables by $(\nu, g, \Psi, Q, \Omega)$ rather than by $(\nu, p_1, p_2, q_1, q_2)$. Using

$$p_1 = g \sin \Psi, \quad p_2 = g \cos \Psi, \quad q_1 = Q \sin \Omega, \quad q_2 = Q \cos \Omega, \tag{81}$$

and the exact GEqOE zonal equations, the first-order osculating dynamics become

$$\dot{\nu} = 0, \tag{82}$$

$$\dot{g} = \frac{U}{c} \beta \sin G = -\frac{A\beta}{cr^3} \left(1 - 3s_i^2 \sin^2 u\right) \sin G, \tag{83}$$

$$\dot{Q} = -\frac{6AQc_i}{cr^3} \sin u \cos u, \tag{84}$$

$$\dot{\Omega} = -\frac{6Ac_i}{cr^3} \sin^2 u, \tag{85}$$

$$\dot{\Psi} = d - w_h - \frac{U}{c} \left(1 + \frac{\cos G}{g}\right) + O(J_2^2). \tag{86}$$

Equations (83)-(86) show the main structural point: once expressed through (G, u), the first-order forcing is a trigonometric-rational function of the anomaly.

The $\dot{\Psi}$ expression must be derived directly from the exact GEqOE p_1, p_2 equations,

$$\dot{\Psi} = \frac{p_2 \dot{p}_1 - p_1 \dot{p}_2}{g^2}, \tag{87}$$

not by shortcut analogy. For the zonal GEqOE right-hand side this gives

$$\dot{\Psi} = d - w_h + \frac{\mathcal{E}_Z}{c} \left(1 + \frac{\cos G}{g}\right), \tag{88}$$

and therefore (86) for J_2 with $\mathcal{E}_Z = -U$.

Multiplying by

$$\frac{dt}{dG} = \frac{r}{w} + O(J_2) \tag{89}$$

gives the anomaly-form equations for the short-period map. For the two magnitude variables:

$$\frac{dg}{dG} = -\frac{A}{\mu r^2} \left(1 - 3s_i^2 \sin^2 u\right) \sin G, \tag{90}$$

$$\frac{dQ}{dG} = -\frac{6AQc_i}{\mu \beta r^2} \sin u \cos u. \tag{91}$$

For the two angles:

$$\frac{d\Omega}{dG} = -\frac{6Ac_i}{\mu \beta r^2} \sin^2 u, \tag{92}$$

$$\frac{d\Psi}{dG} = \frac{r}{w} \dot{\Psi} = \frac{r}{w} \left[d - w_h - \frac{U}{c} \left(1 + \frac{\cos G}{g}\right)\right]. \tag{93}$$

The mean-to-osculating short-period corrections are now obtained by subtracting the secular pieces and integrating over G:

$$\mathcal{G}_1'(G) = \frac{dg}{dG}, \qquad \langle \mathcal{G}_1 \rangle = 0, \tag{94}$$

$$\mathcal{Q}_1'(G) = \frac{dQ}{dG}, \qquad \langle \mathcal{Q}_1 \rangle = 0, \tag{95}$$

$$\mathcal{N}_1'(G) = \frac{d\Omega}{dG} - \frac{r}{w} \omega_{\Omega, J_2}, \qquad \langle \mathcal{N}_1 \rangle = 0, \tag{96}$$

$$\mathcal{P}_1'(G) = \frac{d\Psi}{dG} - \frac{r}{w} \omega_{\Psi, J_2}, \qquad \langle \mathcal{P}_1 \rangle = 0. \tag{97}$$

Hence the first-order map can be written as

$$g_{\rm osc} = \bar{g} + \mathcal{G}_1(\bar{G}) + O(J_2^2), \tag{98}$$
$$Q_{\rm osc} = \bar{Q} + \mathcal{Q}_1(\bar{G}) + O(J_2^2), \tag{99}$$
$$\Psi_{\rm osc} = \bar{\Psi} + \mathcal{P}_1(\bar{G}) + O(J_2^2), \tag{100}$$
$$\Omega_{\rm osc} = \bar{\Omega} + \mathcal{N}_1(\bar{G}) + O(J_2^2), \tag{101}$$

where $\bar{G} = \bar{K} - \bar{\Psi}$.

Finally, the GEqOE components follow immediately:

$$p_{1,\rm osc} = g_{\rm osc} \sin \Psi_{\rm osc}, \tag{102}$$
$$p_{2,\rm osc} = g_{\rm osc} \cos \Psi_{\rm osc}, \tag{103}$$
$$q_{1,\rm osc} = Q_{\rm osc} \sin \Omega_{\rm osc}, \tag{104}$$
$$q_{2,\rm osc} = Q_{\rm osc} \cos \Omega_{\rm osc}. \tag{105}$$

6.4 First-order fast-phase map in L and K

The missing fast-phase piece can be written just as explicitly. Introduce the generalized longitude

$$L = K + p_1 \cos K - p_2 \sin K, \tag{106}$$

so that, for the unperturbed geometry,

$$\frac{\partial L}{\partial K} = 1 - p_1 \sin K - p_2 \cos K = \frac{r}{a}. \tag{107}$$

At first order in J_2 the forward longitude map is

$$L_{\rm osc} = \bar{L} + \ell_1(\bar{\boldsymbol{z}}, \bar{K}) + O(J_2^2), \tag{108}$$

with ℓ_1 normalized by

$$\frac{1}{2\pi} \int_0^{2\pi} \ell_1(\bar{\boldsymbol{z}}, \bar{K})\, d\bar{K} = 0. \tag{109}$$

For the conservative zonal problem, the exact GEqOE L equation reduces at first order to

$$\dot{L} = \nu + d - w_h - \frac{U}{c}\left[\frac{1}{\alpha} + \alpha\left(1 - \frac{r}{a}\right)\right] + O(J_2^2), \tag{110}$$

and therefore

$$\ell_1'(\bar{K}) = \frac{\bar{r}}{\bar{w}}\left[\dot{L}(\bar{\boldsymbol{z}}, \bar{K}) - \langle \dot{L} \rangle_K(\bar{\boldsymbol{z}})\right], \qquad \langle \ell_1 \rangle_K = 0. \tag{111}$$

The important structural point is that ℓ_1 is naturally parameterized by the *absolute* fast angle $\bar{K}$, not by $\bar{G} = \bar{K} - \bar{\Psi}$.

Now expand (106) with the first-order GEqOE map (63)-(64). Writing

$$p_{1,\rm osc} = \bar{p}_1 + u_{p_1}, \qquad p_{2,\rm osc} = \bar{p}_2 + u_{p_2}, \tag{112}$$

one gets

$$\ell_1 = \frac{\bar{r}}{a} v_1 + u_{p_1} \cos \bar{K} - u_{p_2} \sin \bar{K} + O(J_2^2). \tag{113}$$

Using the (g, Ψ) short-period map (101), the component corrections are

$$u_{p_1} = \mathcal{G}_1(\bar{G}) \sin \bar{\Psi} + \bar{g}\, \mathcal{P}_1(\bar{G}) \cos \bar{\Psi}, \tag{114}$$
$$u_{p_2} = \mathcal{G}_1(\bar{G}) \cos \bar{\Psi} - \bar{g}\, \mathcal{P}_1(\bar{G}) \sin \bar{\Psi}. \tag{115}$$

Substituting (114)-(115) into (113) gives the direct fast-phase correction

$$v_1(\bar{\boldsymbol{z}}, \bar{K}) = \frac{a}{\bar{r}} \left[\ell_1(\bar{\boldsymbol{z}}, \bar{K}) + \mathcal{G}_1(\bar{G}) \sin \bar{G} - \bar{g}\, \mathcal{P}_1(\bar{G}) \cos \bar{G} \right]. \tag{116}$$

Equations (108), (111), and (116) are the first-order GEqOE closure for the fast phase. They provide two equivalent practical reconstructions:

1. reconstruct $L_{\text{osc}} = \bar{L} + \ell_1$ and recover K_{osc} from the generalized Kepler relation;
2. reconstruct $K_{\text{osc}} = \bar{K} + v_1$ directly from (116).

The complete first-order mean-element model therefore has three pieces:

1. the secular flow (41)-(54) (or more generally the mean zonal ODE of Section 10);
2. the forward reconstruction map (63)-(64);
3. the inverse initialization map (66)-(67).

The remaining algebraic step is to evaluate the integrals in (97) and (111) in closed elementary form. Because the forcing functions are rational-trigonometric in G (or K), these quadratures can in principle be reduced by the tangent-half-angle substitution. However, the resulting expressions are lengthy, and a more efficient approach is obtained by switching to the complex true anomaly variable.

7 Complex True Anomaly Variables

The short-period integrals of Section 6 are one-revolution quadratures of rational-trigonometric functions in the eccentric anomaly-like variable $G = K - \Psi$. Their symbolic reduction to closed-form expressions is more naturally accomplished by changing variables to the complex true anomaly, following the general philosophy of complex exponential methods in satellite theory [8].

7.1 Definitions

Define the complex true anomaly exponential

$$F = e^{if}, \tag{117}$$

where f is the true anomaly, and the complex argument-of-pericenter exponential

$$w = e^{i\omega}, \qquad \omega = \Psi - \Omega. \tag{118}$$

The eccentricity is encoded in the single parameter

$$q = \frac{g}{1+\beta} = \frac{1-\beta}{g}, \qquad 0 \le q < 1, \tag{119}$$

which satisfies $q \to 0$ in the circular limit ($g \to 0$) and $q \to 1$ in the parabolic limit ($g \to 1$).

The complex true anomaly F is related to the eccentric anomaly exponential $z = e^{iG}$ by the bilinear Möbius transformation

$$F = \frac{z-q}{1-qz}, \qquad z = \frac{F+q}{1+qF}. \tag{120}$$

On unperturbed Keplerian motion, z traces the unit circle $|z| = 1$ as G advances by 2π, and F traces the same unit circle $|F| = 1$ since the Möbius map preserves the unit-circle boundary.

7.2 Mean fast phase and Jacobian

The mean fast phase for the averaged formulation is

$$M = L - \Psi = G - g \sin G, \tag{121}$$

which is exactly the generalized Kepler equation. At first order in the zonal perturbation, M advances uniformly: $\dot{\bar{M}} = \dot{\bar{L}} - \dot{\bar{\Psi}} = \nu + O(\varepsilon)$.

The Jacobian between M and F follows from the chain rule through (121) and (120):

$$\frac{dM}{dF} = -i \, \frac{(1-q^2)^3}{1+q^2} \, \frac{F}{(F+q)^2(1+qF)^2}. \tag{122}$$

This Jacobian has a simple zero at $F = 0$ and double poles at $F = -q$ (inside the unit disk) and $F = -1/q$ (outside the unit disk). Crucially, the denominator $(F+q)^2(1+qF)^2$ is *independent of the zonal degree* n.

7.3 Laurent polynomial structure of the forcing

For a zonal potential of degree n, the Legendre polynomial $P_n(\hat{z})$ with $\hat{z} = s_i \sin u$ is a trigonometric polynomial of degree n in the angle $u = \omega + f$. Through the exponentials $F = e^{if}$ and $w = e^{i\omega}$, the functions $\sin u$, $\cos u$, and their powers become finite Laurent polynomials in F and w.

After separating the w-harmonics, the first-order forcing for each slow variable $v \in \{g, Q, \Psi, \Omega, M\}$ and each zonal degree n decomposes as

$$f_v^{(n)}(F, w) = \sum_m a_{v,m}^{(n)}(F) \, w^m, \tag{123}$$

where each $a_{v,m}^{(n)}(F)$ is a finite Laurent polynomial in F with coefficients in the ring $\mathbb{Q}(q, Q)$.

The parity constraint is:

$$m \equiv n \pmod 2, \tag{124}$$

and the maximum harmonic order is $|m| \leq n - 2$ (not n), as will be established in Section 10. For J_2 ($n = 2$), the only surviving harmonic is $m = 0$; for J_3 ($n = 3$), only $|m| = 1$; and so on.

7.4 Why F is the right variable

The advantage of F over G or K for the short-period integration is threefold:

1. the forcing is a *Laurent polynomial* in F, not a rational-trigonometric function requiring tangent-half-angle substitution;
2. the Jacobian dM/dF has a *fixed* pole structure that does not grow with the zonal degree;
3. the full integrand for each w-harmonic is a rational function of F with known poles, amenable to direct partial-fraction decomposition.

8 Short-Period Kernels: Direct Residue Decomposition

8.1 Homological equation in F

The short-period correction for each slow variable v at zonal degree n satisfies the homological equation

$$\frac{du_1^v}{dM} = f_v^{(n)}(F) - \langle f_v^{(n)} \rangle_M, \tag{125}$$

where $\langle\cdot\rangle_M$ denotes the one-revolution average in M. Converting to integration in F, and separating w-harmonic m:

$$u^v_{1,m}(F) = \int [a_{v,m}(F) - \langle a_{v,m}\rangle_M] \frac{dM}{dF}\, dF. \tag{126}$$

8.2 Integrand structure

Let $a_{v,m}(F) = \sum_{k=k_{\min}}^{k_{\max}} c_k F^k$ be the Laurent polynomial forcing for a given w-harmonic. After clearing negative powers by the shift $s = -k_{\min}$ (when $k_{\min} < 0$), the integrand for (126) takes the universal form

$$\text{scale} \times \frac{N(F)}{F^s\,(F+q)^2\,(1+qF)^2}, \tag{127}$$

where $N(F)$ is a polynomial in F with coefficients in $\mathbb{Q}(q, Q)$, the shift $s \geq 0$ accounts for the original negative F-powers, and the scale factor absorbs the constants from dM/dF.

The denominator structure $(F+q)^2(1+qF)^2$ is inherited directly from the Jacobian (122) and is *independent of the zonal degree* n. Only the polynomial degree of $N(F)$ and the shift s grow with n.

8.3 Algorithm

The direct residue decomposition proceeds in five steps.

Step 1: Polynomial long division. Write

$$N(F) = Q_{\text{poly}}(F) \cdot D(F) + N_{\text{rem}}(F), \tag{128}$$

where $D(F) = F^s(F+q)^2(1+qF)^2$ is the full denominator. The quotient $Q_{\text{poly}}(F)$ is a polynomial of small degree and $\deg N_{\text{rem}} < \deg D$.

Step 2: Double-pole residues at $F = -q$ and $F = -1/q$. The principal parts at the double poles are extracted by evaluation:

$$B = \frac{N_{\text{rem}}(-q)}{(-q)^s\,(1-q^2)^2}, \qquad D_c = \frac{N_{\text{rem}}(-1/q)}{(-1/q)^s\,(-1/q+q)^2}. \tag{129}$$

These give the double-pole contributions to the antiderivative: $-B/(F+q)$ and $-D_c/(q(1+qF))$.

Step 3: Residues at $F = 0$. For $s \geq 2$, the pole at $F = 0$ contributes higher-order residues. The Taylor coefficients χ_j of $\chi(F) = N_{\text{rem}}(F)/[(F+q)^2(1+qF)^2]$ at $F = 0$ are computed iteratively via the convolution identity

$$N_{\text{rem},k} = \sum_{j=0}^{k} D_j\,\chi_{k-j}, \qquad \chi_k = \frac{1}{D_0}\left(N_{\text{rem},k} - \sum_{j=1}^{k} D_j\,\chi_{k-j}\right), \tag{130}$$

where D_j and $N_{\text{rem},k}$ are the Taylor coefficients of $(F+q)^2(1+qF)^2$ and $N_{\text{rem}}(F)$ respectively.

Step 4: Assembly of periodic antiderivative. The antiderivative is assembled from the partial fraction decomposition:

$$u^v_{1,m}(F) = \underbrace{\int Q_{\text{poly}}(F)\,dF}_{\text{polynomial}} - \underbrace{\frac{B}{F+q}}_{\text{pole at } -q} - \underbrace{\frac{D_c}{q(1+qF)}}_{\text{pole at } -1/q} - \underbrace{\sum_{j=2}^{s} \frac{E_j}{(j-1)F^{j-1}}}_{\text{poles at } 0} - C. \tag{131}$$

All logarithmic terms that would arise from simple-pole residues in the standard rational function integration are provably absent. On the physical orbit, F traces the closed contour $|F| = 1$ as M advances by 2π, and the integrand $f_v^{(n)} - \langle f_v^{(n)} \rangle_M$ has zero M-average by construction. Since $\log F$, $\log(F+q)$, and $\log(1+qF)$ are not single-valued on this contour, their coefficients must vanish identically. Formally: the poles $F = -q$ and $F = -1/q$ lie strictly inside and outside the unit circle respectively, so neither contributes a net winding number, and the residue at $F = 0$ is zero because the average has been subtracted.

Step 5: Zero-mean gauge. The constant C is computed from the condition $\langle u_{1,m}^v \rangle_M = 0$ using the structural formulas

$$\left\langle \frac{1}{F+q} \right\rangle_M = \frac{-q(2+q^2)}{(1-q^2)(1+q^2)}, \tag{132}$$

$$\left\langle \frac{1}{1+qF} \right\rangle_M = \frac{1+2q^2}{(1-q^2)(1+q^2)}, \tag{133}$$

$$\left\langle F^k \right\rangle_M = \text{(known closed-form expressions)}. \tag{134}$$

These are derived by substituting $F = (z-q)/(1-qz)$ and evaluating the resulting contour integral at the interior pole $z = 0$.

Computing C term-by-term from (132)–(134) avoids the expression-swell catastrophe that occurs when substituting $F \to F(z)$ into the full rational expression and simplifying.

8.4 Scaling to higher zonal degrees

The method scales from J_2 through J_5 because:

1. the denominator structure $(F+q)^2(1+qF)^2$ is degree-independent;
2. the Laurent polynomial support grows only linearly with n;
3. all operations (long division, evaluation, Taylor recurrence) are polynomial-time in the forcing degree;
4. the final expressions are in canonical single-fraction form, enabling compact serialization.

Three computational optimizations were essential for scaling beyond J_2:

1. *Coefficient-wise scaling*: distributing the scale factor into each F-coefficient individually, rather than multiplying the combined expression, avoids catastrophic multivariate GCD computation in the symbolic algebra system;
2. *Polynomial-ring GCD*: using the specific ring $\mathbb{Q}(q, Q)$ for greatest common divisor computation instead of general-purpose symbolic simplification provides 2–10× speedup;
3. *Cache-aware deserialization*: skipping redundant simplification when loading previously computed expressions from canonical form avoids exponential-time re-simplification.

With these optimizations, the one-time symbolic generation of all J_2–J_5 short-period kernels (5 variables × 4 zonal degrees) completes in approximately 80 minutes of CPU time on a single core.

9 Operational Structure of the Analytical Theory

The preceding sections have derived all the ingredients of a first-order analytical zonal theory for GEqOE. This section summarizes how those ingredients fit together at runtime, and clarifies the sense in which the theory is *analytical* rather than semi-analytical or numerical.

9.1 Precomputation versus runtime

The theory has a clear separation into two phases:

One-time symbolic precomputation (offline). The short-period kernels $\mathcal{G}_1$, $\mathcal{Q}_1$, $\mathcal{P}_1$, $\mathcal{N}_1$, ℓ_1 and the mean drift coefficient functions are derived by the direct residue decomposition of Section 8. This is a symbolic algebra computation that produces exact closed-form rational expressions in the eccentricity parameter q and the inclination parameter Q. The computation is performed once per zonal degree and the resulting expressions are stored. For the present J_2–J_5 theory, this precomputation required approximately 80 minutes of CPU time; the output is a fixed, finite collection of algebraic formulas.

Runtime evaluation (online). Given an osculating GEqOE state, the analytical theory is applied by evaluating the precomputed closed-form expressions at the current orbit parameters. No numerical quadrature, no Fourier transform, no interpolation, and no iterative scheme (apart from the standard generalized Kepler equation) is required. Every operation is the evaluation of an explicit algebraic formula.

This separation is exactly analogous to classical analytical theories such as Brouwer's [2]: the derivation of the short-period and secular expressions is a one-time algebraic effort (historically done by hand, now by computer algebra), while the resulting theory is used by evaluating closed formulas. The present theory is analytical in the same sense.

9.2 Runtime procedure

The complete analytical propagation cycle consists of three steps, each of which involves only the evaluation of closed-form expressions:

Step 1: Osculating-to-mean conversion (initialization). Given an osculating GEqOE state $\boldsymbol{y}_{\text{osc}} = (\nu, p_1, p_2, K, q_1, q_2)_{\text{osc}}$, compute the mean initial state via the first-order inverse map (66)–(67):

$$\bar{\boldsymbol{y}}(t_0) = \boldsymbol{y}_{\text{osc}}(t_0) - \varepsilon\, \boldsymbol{u}_1\big(\boldsymbol{y}_{\text{osc}}(t_0), K_{\text{osc}}(t_0)\big). \tag{135}$$

This requires evaluating the short-period correction functions $\boldsymbol{u}_1$ and v_1 at the initial osculating state—a single pass through the precomputed algebraic formulas.

Step 2: Mean-state propagation. Propagate the mean state $\bar{\boldsymbol{y}}(t)$ forward in time. For the mean state, by construction, all short-period (once-per-orbit) oscillations have been removed. The mean dynamics contain only the secular drift and, for higher zonals, the long-period evolution through the argument of pericenter $\bar{\omega}$.

For pure J_2, this step is trivial: the mean magnitudes $\bar{\nu}$, $\bar{g}$, $\bar{Q}$ are exactly constant, and the mean angles $\bar{\Psi}$, $\bar{\Omega}$ advance linearly in time with the explicit rates (38)–(39). The mean fast phase $\bar{M}$ likewise advances linearly. No differential equation is integrated—the solution is a closed-form function of elapsed time.

For the general J_2–J_5 zonal model, the mean magnitudes $\bar{g}$ and $\bar{Q}$ are no longer exactly constant: they undergo long-period oscillations driven by J_3, J_4, J_5 through the argument of pericenter $\bar{\omega} = \bar{\Psi} - \bar{\Omega}$. The mean-state propagation is then the numerical integration of a low-dimensional autonomous ODE in $(\bar{g}, \bar{Q}, \bar{\Psi}, \bar{\Omega})$ whose right-hand side is a known closed-form function (151)–(154). This ODE is smooth, slowly varying, and can be integrated with large step sizes (tens to hundreds of orbital periods per step), because all fast oscillations have been analytically removed.

It is important to emphasize what this mean-state ODE is *not*: it is not a numerical orbit propagator. It does not resolve individual orbits, does not track the fast angle K, and does

not require step sizes tied to the orbital period. The computational cost is orders of magnitude smaller than that of a full osculating propagator.

Step 3: Mean-to-osculating reconstruction. At any desired output epoch, reconstruct the full osculating state by applying the forward short-period map (63)–(64):

$$\boldsymbol{y}_{\text{osc}}(t) = \bar{\boldsymbol{y}}(t) + \varepsilon\, \boldsymbol{u}_1\big(\bar{\boldsymbol{y}}(t), \bar{K}(t)\big). \tag{136}$$

This again requires only the evaluation of the precomputed short-period formulas at the current mean state. The osculating GEqOE is then converted to Cartesian position and velocity via the standard GEqOE-to-Cartesian map [1], which involves solving the generalized Kepler equation for K.

9.3 Comparison with semi-analytical methods

In the semi-analytical approach (e.g., DSST [12, 13]), the mean-element equations of motion are formulated but the short-period corrections are evaluated by numerical quadrature at each output step, typically requiring one-revolution numerical averages. In the present analytical theory, both the mean drift and the short-period corrections are given by precomputed closed-form expressions. The only numerical component at runtime is the ODE integration of the smooth mean-state system for the general zonal case—and even this disappears entirely for the pure J_2 model.

10 General Zonal Harmonic Structure

10.1 Autonomy is preserved

For a general static zonal potential, the averaged GEqOE system remains autonomous. This is the main reason to attack the zonal problem before adding non-autonomous perturbations. The exact first-order mean system is already given by (29)-(32).

10.2 J_2 is not representative of all zonals

The J_2 closure is unusually simple because the averaged disturbing function depends only on a, e, and i [2]. Higher zonals are not guaranteed to share that property [10, 14]:

1. odd zonals such as J_3 generally retain periapsis dependence in the singly averaged problem;
2. higher even zonals can also introduce more complicated slow-angle structure than the pure J_2 case;
3. the slow subsystem therefore need not remain a uniform rotation in (p_1, p_2).

What does survive is autonomy and axisymmetry. Hence the averaged vector field cannot depend on absolute time and cannot depend on the node angle by itself. In equinoctial variables this implies that the mean zonal dynamics can only depend on:

1. the invariants a, g, and Q;
2. the relative pericenter-orientation information carried by (p_1, p_2) with respect to the symmetry axis encoded by (q_1, q_2).

10.3 General order-N harmonic structure

For any finite truncation

$$U_Z = \sum_{n=2}^{N} U_n, \tag{137}$$

the singly averaged problem remains a *finite-harmonic* system in the single relative angle

$$\bar{\omega} = \bar{\Psi} - \bar{\Omega}. \tag{138}$$

The reason is structural. For a fixed degree n,

$$U_n(\bar{G}, \bar{\omega}) = \frac{C_n}{\bar{r}^{n+1}} P_n(s_i \sin u)\,, \qquad u = \bar{\omega} + f(\bar{G}), \tag{139}$$

with P_n a polynomial of degree n in its argument. Therefore $P_n(s_i \sin u)$ is a finite trigonometric polynomial in u with harmonics no higher than n. After substitution of the rational-trigonometric relations between f and $\bar{G}$, and after one-revolution averaging in $\bar{G}$, the degree-n contribution can only retain harmonics of $\bar{\omega}$ up to order n.

Parity further constrains the result. Because

$$P_n(-x) = (-1)^n P_n(x), \tag{140}$$

the averaged degree-n disturbing function has the form

$$\bar{U}^{(n)}(\bar{\nu}, \bar{g}, \bar{Q}, \bar{\omega}) = \sum_{m \in \mathcal{M}_n} \mathcal{A}_{n,m}(\bar{\nu}, \bar{g}, \bar{Q})\, \Phi_{n,m}(\bar{\omega}), \tag{141}$$

where

$$\mathcal{M}_n = \{m \in \mathbb{N}_0 : m \le n,\ m \equiv n \pmod 2\}, \tag{142}$$

and

$$\Phi_{n,m}(\bar{\omega}) = \begin{cases} \cos(m\bar{\omega}), & n \text{ even}, \\ \sin(m\bar{\omega}), & n \text{ odd}. \end{cases} \tag{143}$$

For even n, the set $\mathcal{M}_n$ includes $m = 0$; for odd n, it does not.

Consequently the averaged GEqOE vector field itself can be written, for any finite zonal

truncation N, as

$$\dot{\bar{g}} = \sum_{\substack{2\le n\le N\\ n \text{ even}}} \sum_{\substack{2\le m\le n\\ m \text{ even}}} \mathcal{G}_{n,m}(\bar{\nu},\bar{g},\bar{Q})\sin(m\bar{\omega}) + \sum_{\substack{2\le n\le N\\ n \text{ odd}}} \sum_{\substack{1\le m\le n\\ m \text{ odd}}} \widetilde{\mathcal{G}}_{n,m}(\bar{\nu},\bar{g},\bar{Q})\cos(m\bar{\omega}), \tag{144}$$

$$\dot{\bar{Q}} = \sum_{\substack{2\le n\le N\\ n \text{ even}}} \sum_{\substack{2\le m\le n\\ m \text{ even}}} \mathcal{Q}_{n,m}(\bar{\nu},\bar{g},\bar{Q})\sin(m\bar{\omega}) + \sum_{\substack{2\le n\le N\\ n \text{ odd}}} \sum_{\substack{1\le m\le n\\ m \text{ odd}}} \widetilde{\mathcal{Q}}_{n,m}(\bar{\nu},\bar{g},\bar{Q})\cos(m\bar{\omega}), \tag{145}$$

$$\dot{\bar{\Psi}} = \sum_{\substack{2\le n\le N\\ n \text{ even}}} \left[\mathcal{P}_{n,0}(\bar{\nu},\bar{g},\bar{Q}) + \sum_{\substack{2\le m\le n\\ m \text{ even}}} \mathcal{P}_{n,m}(\bar{\nu},\bar{g},\bar{Q})\cos(m\bar{\omega})\right] + \sum_{\substack{2\le n\le N\\ n \text{ odd}}} \sum_{\substack{1\le m\le n\\ m \text{ odd}}} \widetilde{\mathcal{P}}_{n,m}(\bar{\nu},\bar{g},\bar{Q})\sin(m\bar{\omega}), \tag{146}$$

$$\dot{\bar{\Omega}} = \sum_{\substack{2\le n\le N\\ n \text{ even}}} \left[\mathcal{N}_{n,0}(\bar{\nu},\bar{g},\bar{Q}) + \sum_{\substack{2\le m\le n\\ m \text{ even}}} \mathcal{N}_{n,m}(\bar{\nu},\bar{g},\bar{Q})\cos(m\bar{\omega})\right] + \sum_{\substack{2\le n\le N\\ n \text{ odd}}} \sum_{\substack{1\le m\le n\\ m \text{ odd}}} \widetilde{\mathcal{N}}_{n,m}(\bar{\nu},\bar{g},\bar{Q})\sin(m\bar{\omega}). \tag{147}$$

10.4 Why these coefficient functions are still analytically reachable

The coefficient functions above are not arbitrary integrals. Under the tangent-half-angle substitution

$$t = \tan\frac{\bar{G}}{2}, \qquad \cos\bar{G} = \frac{1-t^2}{1+t^2}, \qquad \sin\bar{G} = \frac{2t}{1+t^2}, \tag{148}$$

the ellipse geometry becomes rational:

$$\bar{r} = \bar{a}\frac{(1-\bar{g}) + (1+\bar{g})t^2}{1+t^2}, \tag{149}$$

while $\sin f$, $\cos f$, and therefore $\sin u$, $\cos u$, become rational functions of t with coefficients depending only on $(\bar{g},\bar{Q},\bar{\omega})$. Hence every degree-$n$ averaged kernel reduces to a finite rational integral in t.

Equivalently, in the complex true anomaly variable F of Section 7, the coefficient functions arise from contour integrals of rational functions of F on the unit circle, with interior poles only at $F = 0$ and $F = q$ [8]. This is the same residue calculus that underlies the short-period decomposition of Section 8.

10.5 Numerical harmonic probe

A direct probe of the averaged GEqOE drift using the exact zonal fast path already supports (144)-(147). Using the script `docs/geqoe_averaged/zonal_harmonic_probe.py` on a generic eccentric/inclined test orbit gives the dominant slow-angle content summarized in Table 1.

Degree n	Dominant content in $(\dot{\bar{g}}, \dot{\bar{Q}})$	Dominant content in $(\dot{\bar{\Psi}}, \dot{\bar{\Omega}})$
2	0	constant
3	$\cos\bar{\omega}$	$\sin\bar{\omega}$
4	$\sin 2\bar{\omega}$	constant, $\cos 2\bar{\omega}$
5	$\cos\bar{\omega}$, $\cos 3\bar{\omega}$	$\sin\bar{\omega}$, $\sin 3\bar{\omega}$

Table 1: Observed dominant Fourier content of the averaged GEqOE zonal drift for isolated degrees $n = 2, 3, 4, 5$. The numerical least-squares residuals for Fourier bases up to harmonic n are at machine precision for the reported test case, supporting the finite-harmonic structure.

10.6 Maximum harmonic order

The exact degree-n residue formulation reveals a sharper bound than the naive $|m| \leq n$. In the Laurent expansion in $w = e^{i\omega}$, the averaged drift at degree n is first expanded in w-harmonics and only then averaged over the anomaly pole $z = q$. This yields exact finite-sum formulas for the coefficient functions and shows that an *isolated* degree-n zonal term cannot reach harmonic n: the highest surviving slow harmonic is actually $n - 2$. Thus:

$$\begin{aligned} J_{2s+1}: \quad & \dot{\bar{g}}, \dot{\bar{Q}} \sim \sum_{m=1,3,\ldots,2s-1} \cos(m\omega), \qquad \dot{\bar{\Psi}}, \dot{\bar{\Omega}} \sim \sum_{m=1,3,\ldots,2s-1} \sin(m\omega), \\ J_{2s}: \quad & \dot{\bar{g}}, \dot{\bar{Q}} \sim \sum_{m=2,4,\ldots,2s-2} \sin(m\omega), \qquad \dot{\bar{\Psi}}, \dot{\bar{\Omega}} \sim \text{constant} + \sum_{m=2,4,\ldots,2s-2} \cos(m\omega). \end{aligned} \tag{150}$$

10.7 Practical total-order N averaged model

For actual propagation, the degree-by-degree decomposition need not be kept explicit. After summation over $n = 2, \ldots, N$, the total averaged slow system can be written directly in the reduced finite basis

$$\dot{\bar{g}} = \sum_{\substack{1 \leq m \leq N \\ m \text{ odd}}} G_m^{(c)}(\bar{\nu}, \bar{g}, \bar{Q}) \cos(m\bar{\omega}) + \sum_{\substack{2 \leq m \leq N \\ m \text{ even}}} G_m^{(s)}(\bar{\nu}, \bar{g}, \bar{Q}) \sin(m\bar{\omega}), \tag{151}$$

$$\dot{\bar{Q}} = \sum_{\substack{1 \leq m \leq N \\ m \text{ odd}}} Q_m^{(c)}(\bar{\nu}, \bar{g}, \bar{Q}) \cos(m\bar{\omega}) + \sum_{\substack{2 \leq m \leq N \\ m \text{ even}}} Q_m^{(s)}(\bar{\nu}, \bar{g}, \bar{Q}) \sin(m\bar{\omega}), \tag{152}$$

$$\dot{\bar{\Psi}} = P_0(\bar{\nu}, \bar{g}, \bar{Q}) + \sum_{\substack{1 \leq m \leq N \\ m \text{ odd}}} P_m^{(s)}(\bar{\nu}, \bar{g}, \bar{Q}) \sin(m\bar{\omega}) + \sum_{\substack{2 \leq m \leq N \\ m \text{ even}}} P_m^{(c)}(\bar{\nu}, \bar{g}, \bar{Q}) \cos(m\bar{\omega}), \tag{153}$$

$$\dot{\bar{\Omega}} = N_0(\bar{\nu}, \bar{g}, \bar{Q}) + \sum_{\substack{1 \leq m \leq N \\ m \text{ odd}}} N_m^{(s)}(\bar{\nu}, \bar{g}, \bar{Q}) \sin(m\bar{\omega}) + \sum_{\substack{2 \leq m \leq N \\ m \text{ even}}} N_m^{(c)}(\bar{\nu}, \bar{g}, \bar{Q}) \cos(m\bar{\omega}). \tag{154}$$

This is the most useful statement for a reusable GEqOE averaged theory: the entire order-N zonal model is parameterized by a finite set of coefficient functions of $(\bar{\nu}, \bar{g}, \bar{Q})$ only.

These coefficient functions are obtained by standard Fourier projection in $\bar{\omega}$. For example,

$$G_m^{(c)} = \frac{1}{\pi}\int_0^{2\pi} \dot{\bar{g}}(\bar{\nu},\bar{g},\bar{Q},\bar{\omega})\cos(m\bar{\omega})\,d\bar{\omega}, \qquad m \text{ odd}, \tag{155}$$

$$G_m^{(s)} = \frac{1}{\pi}\int_0^{2\pi} \dot{\bar{g}}(\bar{\nu},\bar{g},\bar{Q},\bar{\omega})\sin(m\bar{\omega})\,d\bar{\omega}, \qquad m \text{ even}, \tag{156}$$

$$P_0 = \frac{1}{2\pi}\int_0^{2\pi} \dot{\bar{\Psi}}(\bar{\nu},\bar{g},\bar{Q},\bar{\omega})\,d\bar{\omega}, \tag{157}$$

$$P_m^{(s)} = \frac{1}{\pi}\int_0^{2\pi} \dot{\bar{\Psi}}(\bar{\nu},\bar{g},\bar{Q},\bar{\omega})\sin(m\bar{\omega})\,d\bar{\omega}, \qquad m \text{ odd}, \tag{158}$$

$$P_m^{(c)} = \frac{1}{\pi}\int_0^{2\pi} \dot{\bar{\Psi}}(\bar{\nu},\bar{g},\bar{Q},\bar{\omega})\cos(m\bar{\omega})\,d\bar{\omega}, \qquad m \text{ even}, \tag{159}$$

and analogously for the Q and Ω equations.

Numerically, this reduced total-order basis works not only degree by degree, but also after summing the harmonics. In the current probe, the combined $J_2 + J_3 + J_4 + J_5$ averaged drift is reproduced by (151)-(154) with relative least-squares residuals at the level of 10^{-12} to 10^{-14}.

10.8 Explicit symbolic results

The symbolic route has been carried through J_5. A companion note, `docs/geqoe_averaged/zonal_symbolic_co` records closed formulas for the isolated J_3 and J_4 contributions. In particular:

1. the J_3 averaged drift closes exactly as

$$\dot{\bar{g}}_{J_3}, \dot{\bar{Q}}_{J_3} \propto \cos\omega, \qquad \dot{\bar{\Psi}}_{J_3}, \dot{\bar{\Omega}}_{J_3} \propto \sin\omega; \tag{160}$$

2. the J_4 averaged drift closes exactly as

$$\dot{\bar{g}}_{J_4}, \dot{\bar{Q}}_{J_4} \propto \sin(2\omega), \qquad \dot{\bar{\Psi}}_{J_4}, \dot{\bar{\Omega}}_{J_4} = \text{constant} + \propto \cos(2\omega). \tag{161}$$

Those explicit formulas were checked against the exact averaged zonal fast path and match to numerical precision in the current validation suite.

The general degree-n generator is recorded in companion note `docs/geqoe_averaged/zonal_symbolic_gene` which includes the exact degree-n generator in closed combinatorial and derivative form, and the explicit J_5 formulas generated from that machinery.

10.9 Validation against numerical zonal propagation

The symbolic mean drift model has been validated against numerical propagation at the *slow-flow* level, as recorded in companion note `docs/geqoe_averaged/zonal_mean_validation.tex`. The validation checks:

1. pointwise parity between the exact symbolic averaged RHS and the numerical one-revolution average of the full zonal RHS;

2. long-arc comparison between the propagated exact averaged model and per-orbit means extracted from the full osculating zonal solution.

For the reference case

$$a = 16000 \text{ km}, \qquad e = 0.35, \qquad i = 50^\circ, \qquad \Omega_0 = 25^\circ, \qquad \omega_0 = 40^\circ, \qquad M_0 = 30^\circ,$$

the long-arc slow-flow agreement over 30 orbital periods is:

$$\mathrm{RMS}_{\mathrm{rel}}(p_1, p_2, q_1, q_2) \approx (2.1,\ 6.3,\ 6.4,\ 1.6) \times 10^{-6}, \qquad \mathrm{RMS}(\Psi, \Omega) \approx (1.6,\ 3.1) \times 10^{-6}\ \mathrm{rad}. \tag{162}$$

The remaining pointwise parity gap against the *full* numerical one-revolution average is small but nonzero for $\dot{\bar{Q}}, \dot{\bar{\Psi}}, \dot{\bar{\Omega}}$; importantly, it decreases approximately linearly when all zonals are scaled down, which is the expected signature of an $O(J^2)$ remainder in a first-order averaged theory rather than an incorrect first-order closure.

11 Numerical Validation of the Short-Period Map

The complete mean-plus-short-period model is validated against full osculating GEqOE Taylor integration with the zonal potential $\{J_2, J_3, J_4, J_5\}$.

11.1 Test orbits

Two representative test orbits are used:

Case	a [km]	e	i [deg]	Ω_0 [deg]	ω_0 [deg]
low-e	9000	0.05	40	25	60
high-e	18000	0.65	63	40	250

The low-eccentricity case represents near-circular orbits where the short-period corrections are small (the eccentricity parameter $q \approx 0.025$ places the poles far from the unit circle); the high-eccentricity case ($q \approx 0.44$) stresses the pole structure and tests the convergence of the first-order theory.

11.2 Reconstruction procedure

For each test orbit and multiple initial mean anomalies $M_0 \in \{20, 80, 140, 220, 300\}^\circ$ (low-e) and $M_0 \in \{35, 90, 170, 260, 340\}^\circ$ (high-e), the following procedure is applied:

1. convert the osculating initial state to mean elements via the first-order inverse map (66)-(67);
2. propagate the mean state for 10 (low-e) or 8 (high-e) orbital periods using the exact symbolic mean drift;
3. at each output epoch, apply the forward short-period map (63)-(64) to reconstruct the osculating GEqOE state;
4. solve the generalized Kepler equation for K and convert to Cartesian position;
5. compare against the reference osculating Taylor integration.

11.3 J_2–J_5 multi-anomaly results

Table 2 summarizes the reconstruction errors.

The low-eccentricity case achieves sub-300-meter RMS position accuracy; the high-eccentricity case stays below 4 km RMS. The results are indistinguishable from J_2–J_4, confirming that J_5 contributes negligibly to the first-order short-period corrections.

Case	M_0	K RMS [rad]	pos RMS [km]	pos max [km]	Ψ RMS [rad]
low-e	20°	9.51e-6	0.155	0.297	1.22e-5
low-e	80°	7.74e-6	0.189	0.426	1.23e-5
low-e	140°	8.92e-6	0.164	0.344	1.21e-5
low-e	220°	1.58e-5	0.255	0.517	1.64e-5
low-e	300°	1.79e-5	0.299	0.581	2.67e-5
high-e	35°	2.41e-4	2.909	4.568	2.42e-4
high-e	90°	1.91e-4	2.235	3.840	1.91e-4
high-e	170°	1.35e-4	1.823	3.143	1.35e-4
high-e	260°	1.83e-4	3.025	5.661	1.83e-4
high-e	340°	2.32e-4	3.820	6.852	2.31e-4

Table 2: Osculating reconstruction errors for the J_2–J_5 mixed-zonal model across multiple initial anomalies.

11.4 Inverse round-trip test

The internal consistency of the first-order map is tested by the cycle $\boldsymbol{y}_{\rm osc} \to \bar{\boldsymbol{y}} \to \boldsymbol{y}_{\rm rec}$:

Case	Cartesian round-trip error
low-e	7.2 m
high-e	1.6 m

The sub-10-meter round-trip error confirms that the forward and inverse maps are self-consistent to well within the first-order accuracy of the theory.

11.5 Zonal scaling test

To verify that the reconstruction error is dominated by the expected $O(\varepsilon^2)$ residual of a first-order theory, the validation is repeated with the zonal coefficients scaled by factors $\lambda \in \{1.00, 0.50, 0.25, 0.125\}$. A log-log fit gives

$$\frac{d\log(\mathrm{RMS}_{\rm pos})}{d\log\lambda} = 0.998. \tag{163}$$

The near-unity slope confirms that the reconstruction error scales linearly with the perturbation parameter ε, consistent with the $O(\varepsilon)$ short-period signature being the dominant error source. A second-order truncation artifact would produce a slope of approximately 2.

11.6 Anomaly and eccentricity dependence

Two features of Table 2 merit discussion.

Anomaly dependence. The reconstruction error varies by roughly 2× across initial anomalies for both test orbits. This arises because the first-order inverse map $\bar{\boldsymbol{y}}(t_0) = \boldsymbol{y}_{\rm osc}(t_0) - \varepsilon\,\boldsymbol{u}_1(\boldsymbol{y}_{\rm osc}(t_0), K_{\rm osc}(t_0))$ commits an $O(\varepsilon^2)$ initialization error whose magnitude depends on where on the orbit the initial condition falls. Near periapsis, where the short-period correction has its steepest gradient, the quadratic remainder is largest. The effect is intrinsic to any first-order near-identity map and would be removed by a second-order theory.

Eccentricity dependence. The high-eccentricity case shows roughly 15× larger errors than the low-eccentricity case. Three mechanisms contribute:

1. *Pole proximity*: the eccentricity parameter q controls the distance from the integration contour $|F| = 1$ to the poles at $F = -q$ and $F = -1/q$. As $q \to 1$, these poles approach the unit circle and the short-period corrections develop large amplitudes, making the $O(\varepsilon^2)$ neglected terms proportionally larger.

2. *Periapsis concentration*: the orbital velocity peaks sharply near periapsis for eccentric orbits, concentrating the short-period variation into a small fraction of the orbital period and amplifying the Cartesian position error per unit GEqOE error.

3. *Kepler equation sensitivity*: the mapping from GEqOE to Cartesian coordinates involves solving the generalized Kepler equation, whose sensitivity to small GEqOE errors grows with eccentricity.

11.7 Validation figures

The component-level reconstruction is shown in Figure 1 (low-e) and Figure 2 (high-e). The Cartesian position errors across the anomaly sweep are collected in Figure 3, and the zonal scaling test is shown in Figure 4.

12 Path to a Fourier-in-K Thrust Extension

Consider adding a small non-conservative acceleration expressed in the orbital RTN frame,

$$\boldsymbol{P}(K) = P_r(K)\,\boldsymbol{e}_r + P_t(K)\,\boldsymbol{e}_t + P_h(K)\,\boldsymbol{e}_h, \tag{164}$$

with Fourier expansions in the propagated GEqOE fast angle K,

$$P_r(K) = a_{r0} + \sum_{m=1}^{M} \left(a_{rm}\cos mK + b_{rm}\sin mK\right), \tag{165}$$

$$P_t(K) = a_{t0} + \sum_{m=1}^{M} \left(a_{tm}\cos mK + b_{tm}\sin mK\right), \tag{166}$$

$$P_h(K) = a_{h0} + \sum_{m=1}^{M} \left(a_{hm}\cos mK + b_{hm}\sin mK\right). \tag{167}$$

Let $\boldsymbol{c}$ denote the stacked Fourier coefficients. If one linearizes the forced averaged problem about the autonomous zonal slow flow and neglects the mixed zonal-thrust coupling at first pass, the mean dynamics take the form

$$\dot{\bar{\boldsymbol{z}}} = \boldsymbol{f}^{sec}_{\text{zonal}}(\bar{\boldsymbol{z}}) + \boldsymbol{B}(\bar{\boldsymbol{z}})\,\boldsymbol{c} + O(J_n\|\boldsymbol{c}\|, \|\boldsymbol{c}\|^2), \tag{168}$$

where $\boldsymbol{f}^{sec}_{\text{zonal}}$ is the autonomous slow flow.

The main structural reason this is promising is that in GEqOE:

1. r/a is affine in $\sin K$ and $\cos K$;

2. X/a and Y/a are affine in $\sin K$ and $\cos K$;

3. the averaging weight is also r/a.

Therefore the thrust-induced averaged kernels are low-order trigonometric polynomials in K. Before retaining zonal-thrust coupling, the resulting matrix $\boldsymbol{B}(\bar{\boldsymbol{z}})$ should depend only on a finite low-order subset of the Fourier coefficients. This is the GEqOE analogue of the classical thrust Fourier coefficient reduction, and it is the correct next target once the autonomous zonal baseline is fully controlled.

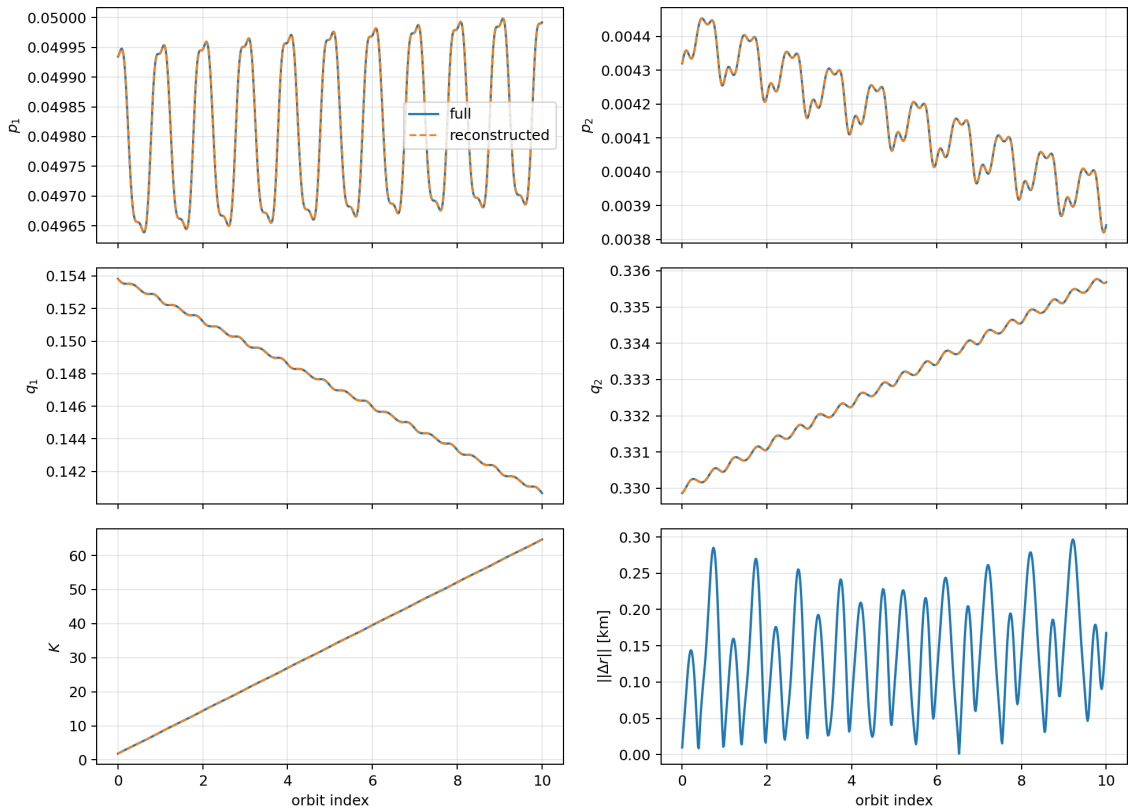

Figure 1: Low-eccentricity osculating GEqOE and Cartesian reconstruction. Solid: reference osculating Taylor integration. Dashed: mean-plus-short-period reconstruction. Bottom panel: Cartesian position error.

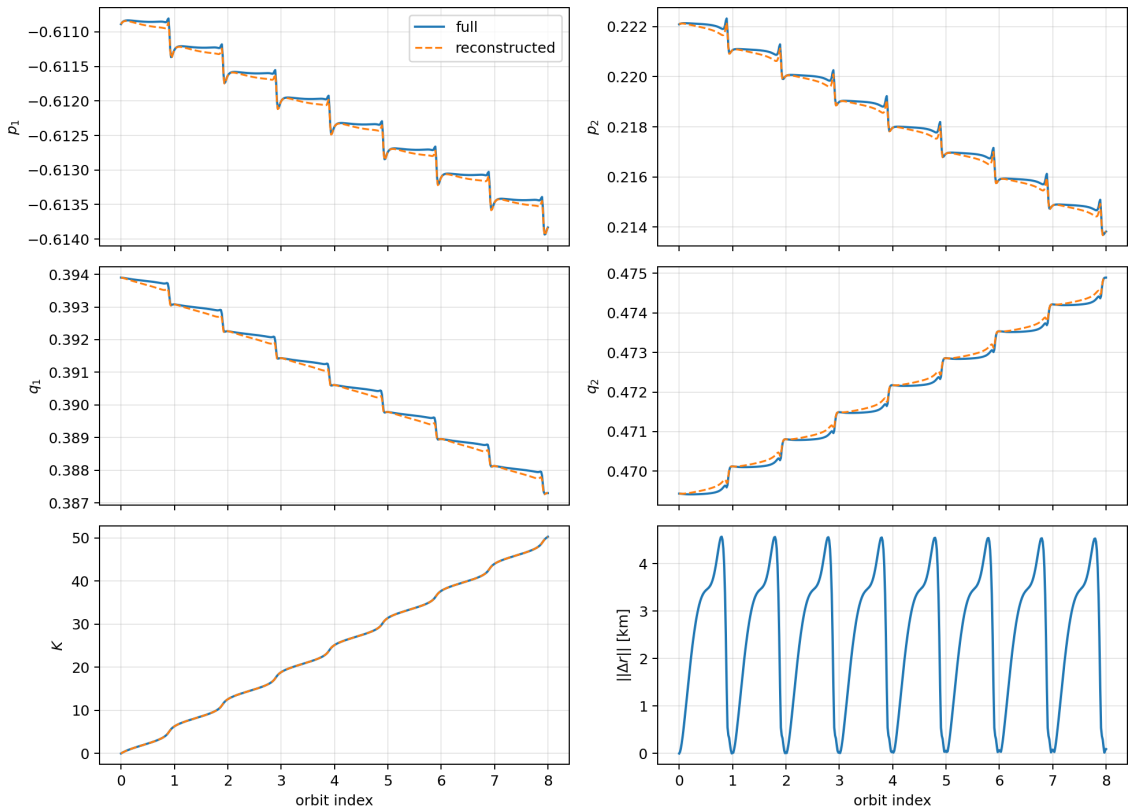

Figure 2: High-eccentricity osculating GEqOE and Cartesian reconstruction.

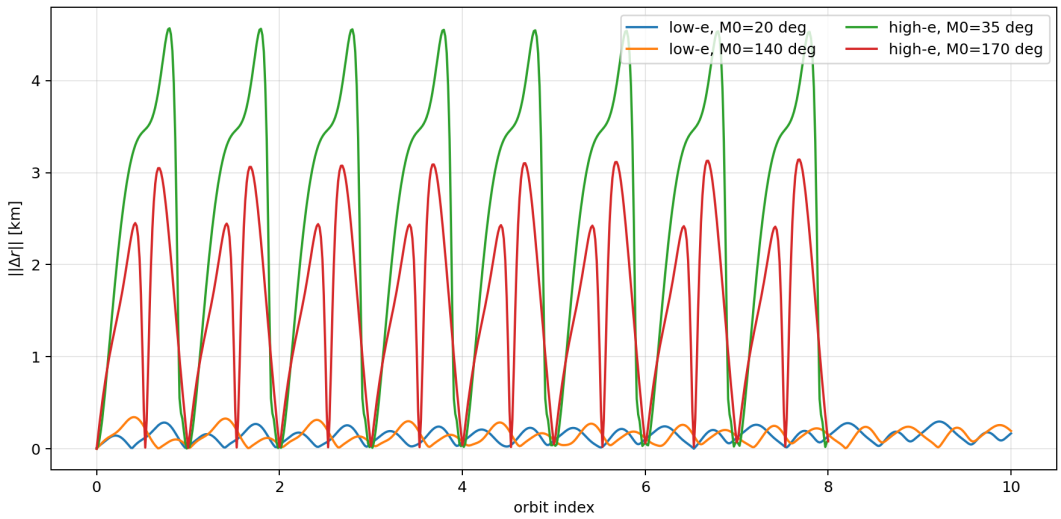

Figure 3: Cartesian position reconstruction error across the anomaly sweep for both test orbits.

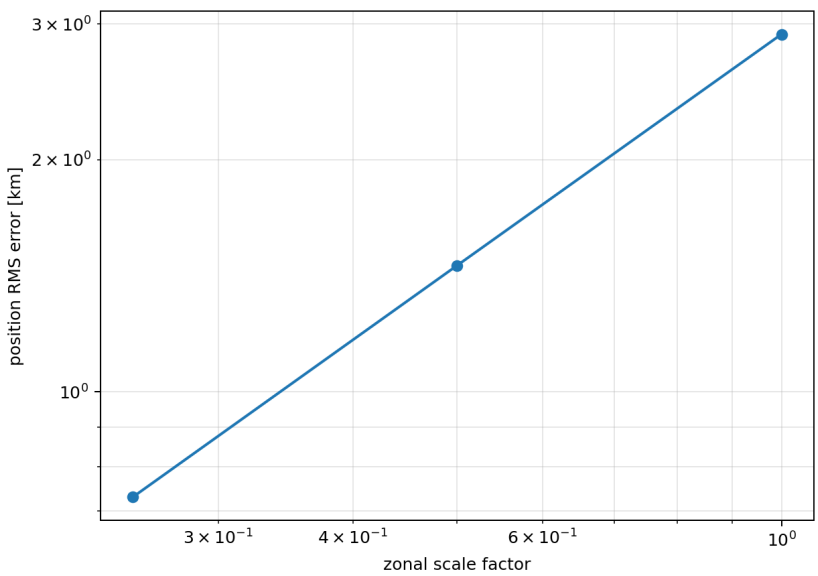

Figure 4: RMS Cartesian reconstruction error versus zonal scale factor. The fitted log-log slope of 0.998 confirms the expected first-order scaling.

13 Conclusions

This note has presented a complete first-order near-identity averaging formulation for GEqOE under autonomous Earth zonal harmonics through J_5. The main contributions are:

1. The GEqOE state vector admits exact first-order orbit averaging in the generalized eccentric longitude K, with the averaged slow-angle dependence reducing to a finite Fourier series in the single relative angle $\omega = \Psi - \Omega$. For pure J_2, the secular closure is especially simple: the magnitudes g and Q are constant and the equinoctial pairs undergo uniform rotations with the explicit rates (38)–(39). For general zonal truncation order N, the mean system is a finite-harmonic ODE parameterized by coefficient functions of $(\bar{\nu}, \bar{g}, \bar{Q})$ alone.

2. A direct residue decomposition method enables the symbolic construction of all short-period kernels. By exploiting the fixed pole structure of the dM/dF Jacobian, the method avoids the logarithmic-term and expression-swell pathologies of standard rational function integration. All J_2–J_5 kernels for the five slow variables (g, Q, Ψ, Ω, M) have been computed and validated, with one-time symbolic generation completing in approximately 80 minutes.

3. Numerical validation against full osculating Taylor integration demonstrates sub-kilometer position reconstruction for low-eccentricity orbits and few-kilometer reconstruction for high-eccentricity ($e = 0.65$) orbits over 10+ orbital periods, with the reconstruction error scaling linearly with the zonal perturbation parameter (slope = 0.998) as expected for a first-order theory. The inverse round-trip error is sub-10 meters, confirming internal consistency.

The formulation is now closed at first order for the autonomous zonal problem. The natural extensions are:

1. second-order short-period corrections to reduce the reconstruction error for high-eccentricity orbits;

2. extension of the short-period map to second order, which would also require the second-order mean drift corrections from cross-zonal products $O(J_n J_m)$;

3. a Fourier-in-K thrust averaging layer built on top of the zonal baseline, as sketched in Section 12.

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